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Optimization of lap-joint laser welding parameters using high-fidelity simulations and machine learning mode

Yung-An Tsai ^a, Yu-Lung Lo ^{a,b,*}, M. Mohsin Raza ^a, Ali N. Saleh ^a,
Tzu-Ching Chuang ^a, Cheng-Yen Chen ^c, Chi-Pin Chiu ^c

^a Department of Mechanical Engineering, National Cheng Kung University, Tainan, Taiwan

^b Academy of Engineering, National Cheng Kung University, Tainan, Taiwan

^c Jum-bo Co., Ltd, Tainan, Taiwan

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ABSTRACT

In lap joint laser welding, a common practice is to conduct trial-and-error experiments using various parameter settings to determine processing conditions that enhance the quality of the weld. However, these experiments are both time-consuming and expensive. Therefore, in this study, we propose a more systematic approach for determining the optimal laser power and scanning speed in the lap joint of SS316 by using highly accurate simulations and artificial neural network models. The processing maps were obtained for three criteria: the melt pool depth, melt pool width, and cooling rate, respectively, which were screened using appropriate quality criteria to determine the laser power and scanning speed that could simultaneously minimize porosity, the size of the heat affected zone, and residual stress. The validity of the simulation model was confirmed by comparing the simulation results of the melt pool geometry with the experimental data. The mean deviations of the experimental and simulated results for melt pool depth and width were found to be only 5.34% and 10%, respectively. As a result, the joint welds produced using the optimal processing parameters exhibited minimal porosity, which was reduced from 1.22% in a non-penetration zone to 0.21% in an optimized zone. Additionally, these welds achieved an ultimate shear strength of up to 545.77 MPa, which is approximately 32% higher than that of the original base metal. Therefore, the effectiveness of the proposed framework for determining the optimal processing conditions for joint laser welding of SS316 has been confirmed.

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1. Introduction

Laser welding has many practical advantages over traditional welding techniques, including high quality high accuracy, excellent quality, high speed, ease of processing, and low

distortion. Furthermore, laser welding is readily integrated with robot technology to realize fully automated welding operations for production line applications or hostile working environments. However, the laser processing parameters, e.g., the laser power and scanning speed, have a critical effect on the

* Corresponding author.

E-mail address: loyl@mail.ncku.edu.tw (Y.-L. Lo).

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temperature field within the weldment and must therefore be carefully chosen to ensure the quality of the welding outcome [1]. Laser weldments are prone to many defects, such as distortion, porosity, and vulnerable surface quality, and these defects have a serious effect on the mechanical performance and surface integrity of the welded part [2]. The interfacial bonding strength between different materials is crucial for many engineering applications. The presence of intermetallic compounds (IMCs) and metallurgical reactions at the interface can enhance bonding strength, while lattice mismatch, microdefects, residual stress, and crystal defects can negatively impact bonding strength [3]. The processing conditions required to minimize these defects are generally determined using experimental trial-and-error methods. However, the processing parameters have both individual and interactive effects on the welding outcome. Consequently, trial-and-error approaches are not only time-consuming and expensive, but may also fail to determine the optimal outcome. Consequently, more systematic and reliable methods for determining the appropriate processing conditions are required.

High-power laser welding has a complex heat and mass transfer behavior, and weld defects such as undercutting, spatter, and humping readily occur. Thus, many experimental studies have been conducted to examine the dynamic behavior of the keyhole and melt pool in high-power laser deep-penetration welding. Kawahito et al. [4] used a 10 kW fiber laser to evaluate the effects of the laser power, power density, and welding speed on the quality of bead-on-plate SS304 stainless steel welds. The welding process was observed using a high-speed video camera and an X-ray transmission real-time imaging system. However, the experimental observations provided no insights into the internal reaction between the molten weld pool and the keyhole. Thus, in recent years, numerical simulation methods have gained increasing interest as a means of exploring the complicated physical phenomena associated with laser welding, such as the dynamics of the melt pool and keyhole, respectively [5], the weld bead appearance [6], the formation of porosity [7], the generation of spatter [8], and so on.

Accordingly, various researchers have used more advanced heat source models in conjunction with ray-tracing algorithms to describe the reflection behavior of the laser beam within the keyhole more precisely. For example, Cho et al. [9] combined a volume of fluid (VOF) method with a ray-tracing technique to analyze the dynamic behaviors of the keyhole and melt pool in the high-power disk laser welding process. Lin et al. [10] investigated the keyhole dynamics and porosity formation in the remote laser welding of aluminum alloy. Panwisawas et al. [11] used a physics-based model based on a comprehensive consideration of the heat transfer, fluid flow, and internal reaction phenomena in the melt pool to simulate the keyhole formation and porosity generation process in the laser welding of Ti6Al4V titanium alloy. Similarly, Pang et al. [12] developed a novel 3D transient multiphase model based on the Marangoni shear stress, Fresnel absorption, heat transfer, fluid flow, and phase transformation phenomena for investigating the keyhole, vapor plume, and weld pool dynamics in deep-penetration fiber laser welding. The simulation results for the weld bead dimensions, transient keyhole instability, weld pool evolution, and vapor

plume behavior were all shown to be in good agreements with the experimental results. However, while the studies in Refs. [9–12] demonstrate the feasibility for utilizing physics models to investigate the melt pool dynamics and keyhole behavior in deep-penetration laser welding, few studies have attempted to utilize such models to determine the optimal processing conditions for the laser beam welding of SS316 stainless steel.

Accordingly, the present study commences by constructing a high-fidelity 3D computational fluid dynamics (CFD) model to analyze the dynamic behavior of the keyhole and melt pool in the single-track laser processing of SS316 substrates using various laser powers and scanning speeds. To ensure the accuracy of the simulation results, the model takes account not only of the Fresnel absorption effect, but also the recoil pressure, Marangoni force, buoyancy force, and vapor-induced plume heat transfer. The validity of the numerical model is confirmed by comparing the simulation results for the dimensions of the melt pool under representative processing conditions with the experimental observations. The simulation model is then used to construct an artificial neural network (ANN) for predicting the melt pool depth, melt pool width, and cooling rate for any combination of the laser processing parameters within the feasible design space. Finally, the processing maps produced by the ANN are screened by three quality criteria to determine the processing parameters (i.e., laser power and scanning speed) which minimize the residual stress and porosity of the lap-joint welds and improve their mechanical properties. The viability of the selected parameters is confirmed by performing experimental tensile tests with a universal testing machine. It is noted that high-fidelity numerical simulation method in this study can offer accurate predictions of the welding process by capturing the complex physics of the process. By understanding the formation mechanism of defects such as the porosity, the heat affected zone (HAZ), and residual stress, the strategies can be developed to mitigate or prevent them. Therefore, the high-fidelity numerical simulation method is a valuable tool for studying the optimal parameters of the laser welding and thus has several advantages over other methods.

2. Methodology

2.1. Assumptions and governing equations

As shown in Fig. 1, the CFD simulations considered the single-track lap-joint laser welding of two SS316 plates with dimensions of $10 \times 5 \times 0.2$ cm. For computational expediency, the computational domain for the CFD simulations was restricted to a 3D cube with dimensions of $4.5 \times 3 \times 4$ mm. The aim of the simulation process was to predict the melt pool dimensions and cooling rate for any given values of the main laser welding parameters (i.e., the laser power and scanning speed).

The main assumptions and governing equations used in developing the CFD model are described in the following.

- (1) The molten metal in the weld pool undergoes incompressible laminar flow has Newtonian viscosity.

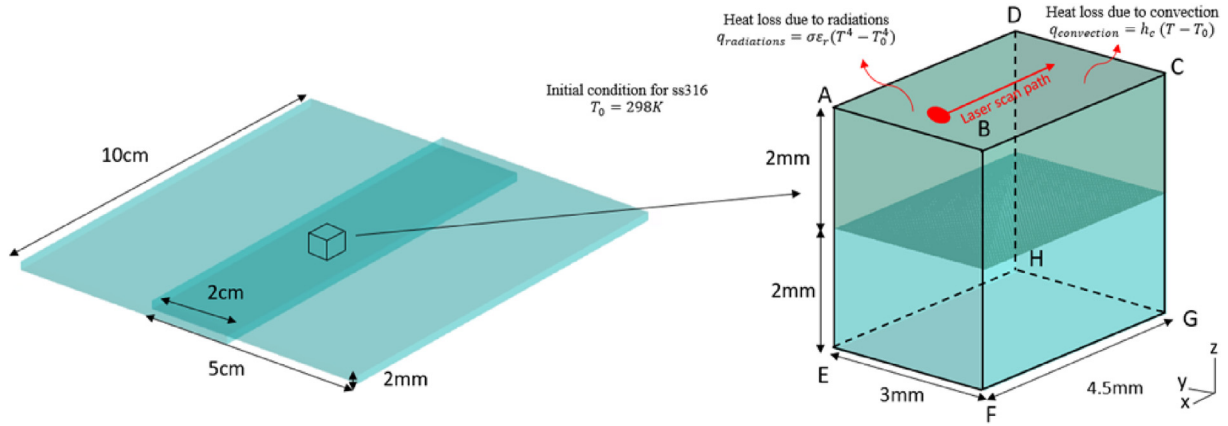


Fig. 1 – Lap-joint laser welding simulation model.

- (2) Solid or liquid steel is denser than gas or vapor, so the gas or vapor plume region is void [13]. Furthermore, the bubbles in the melt pool are adiabatic bubbles, the laser was modeled with Flow3D software.
- (3) The energy density of the laser beam has a Gaussian distribution.
- (4) The recoil pressure induced by evaporation can be determined by the Clausius-Clapeyron equation [14]. The laser beam profile is assumed to be characterized by the type of Gaussian-like axisymmetric distribution. The heat source in the CFD simulations is modeled as [10];

$$q = Q \cdot \exp \left[- \left(\frac{r}{r_b} \right)^2 \right] \quad (1)$$

where q is the heat density function, Q is the heat density value at the beam center, r is the radial coordinate, and r_b is the laser beam radius at the focal plane. The paths of the light rays emitted by the laser were modeled using a ray-tracing technique based on the real-time multiple reflection model in Flow3D®. Moreover, the volume of fluid (VOF) method was used to track the movement of the free surface of the keyhole during the welding process.

The mass conservation equation is formulated as [9].

$$\nabla v = \frac{\dot{m}_s}{\rho} \quad (2)$$

where v is the velocity vector and $\dot{m}_s$ is the mass source term contributed by molten drops of metal.

The momentum conservation equation has the form as [9].

$$\frac{\partial v}{\partial t} + v \cdot \nabla v = - \frac{1}{\rho} \nabla p + \mu \nabla^2 v - K v + \frac{\dot{m}_s}{\rho} (v_s - v) + G \quad (3)$$

where v is the velocity vector, v_s is the velocity vector for the mass source, ρ is the fluid density, p is the pressure, μ is the dynamic viscosity, and G is the body acceleration force. In addition, K is the drag coefficient for a porous media model in the mushy zone and is defined as $K = C(F_s^2 / ((1 - F_s)^3 + B))$, where C is a constant representing the mushy zone

morphology, F_s is the solid fraction, and B is a positive non-zero value use to avoid division by zero.

The energy conservation equation is formulated as [9].

$$\frac{\partial h}{\partial t} + v \cdot \nabla h = \frac{1}{\rho} \cdot (\nabla \cdot (k \nabla T)) \quad (4)$$

where h is the enthalpy and T is the temperature. It is noted that the body force is neglected in Eq. (4) in this study. The solid-liquid phase transition is described using the following enthalpy-based continuum model as [9].

$$h = \begin{cases} \rho_s C_s T & (T \leq T_s) \\ h(T_s) + h_{sl} \frac{T - T_s}{T_l - T_s} & (T_s \leq T \leq T_l) \\ h(T_l) + \rho_l C_l (T - T_l) & (T_l \leq T) \end{cases} \quad (5)$$

where ρ_s and ρ_l are the solid and liquid densities, C_s and C_l are the specific heats per unit volume of the solid and liquid phases, T_s and T_l are the solidus and liquidus temperatures, and h_{sl} is the latent heat of fusion. The volume of fluid (VOF) equation has the form as [9]

$$\frac{\partial F}{\partial t} + \nabla \cdot (vF) = \dot{F}_s \quad (6)$$

where F is the volume of fluid and $\dot{F}_s$ is the change of the volume fraction of fluid associated with the mass source term $\dot{m}_s$ in the mass continuity equation.

Keyhole formation is of significant interest in the laser welding process and is governed mainly by the recoil pressure produced by the evaporation of molten metal from the weld pool. The recoil pressure, P_R , acting on the keyhole wall can be formulated as [9].

$$P_R(T) \cong 0.54 P_{sat}(T) \cong 0.54 P_0 \exp \frac{L_v}{R T} \frac{T - T_b}{T_b} \quad (7)$$

where P_{sat} is the saturated pressure, P_0 is the atmospheric pressure, and L_v is the latent heat of vaporization. Furthermore, T_b is the boiling temperature, and R is the universal gas constant. The recoil pressure described in Eq. (8) leads to a depression of the melt pool when the temperature reaches the evaporation point. It is noted that the liquid metal's surface

Table 1 – Thermal properties used in simulations [18,19].

Parameters	Values
Density (kg/m ³)	7950 – 0.5(T – 298); 300K < T _L ≤ 1723K 6881 – 0.77(T – 1723); 1723K < T _v ≤ 3375K
Solidus temperature T _S (K)	1658
Liquidus temperature T _L (K)	1723
Evaporation temperature T _v (K)	3200
Recrystallization temperature T _R (K)	1223
Specific heat (J/kg·K)	470 + 0.184(T – 300); T _L ≤ 1723K 830; 1723K < T _v ≤ 3375K
Thermal conductivity	13.4 + 0.0136(T – 300); 300K < T _L ≤ 1723K 28.5 + 0.00886(T – 1723); 1723K < T _v ≤ 3375K
Latent heat of fusion (J/kg)	2.6 × 10 ⁵
Latent heat of evaporation (J/kg)	7.4 × 10 ⁶
Dynamic viscosity (kg/m·s)	0.0056; T _L ≤ 1723K 0.0056 – 3.721 × 10 ⁻⁶ (T – 1723K); 1723K < T _v ≤ 3200K
Surface tension (N/m)	3.28
Temperature coefficient of surface tension N/(m·K)	–0.00089
Initial temperature of base metal (K)	298
Laser beam diameter (μm)	50

tension and generates a surface tension force which varies as the following linear function of the temperature as [9]

$$\gamma = \gamma_0 + \frac{d\gamma}{dT}(T - T_0) \quad (8)$$

where γ is the surface tension and γ_0 is the surface tension of the corresponding pure metal at its melting point. In this study, the role of surface-active elements was neglected in the weld pool and the $\frac{d\gamma}{dT}$ term is negative, and liquid metal flows outwards from the heat source along the surface of the weld pool [9].

Finally, the Fresnel absorptivity $\alpha(\epsilon, \theta)$ of the laser energy within the keyhole is a function of the individual ray incident angle (θ) with respect to the surface normal and the electrical conductance (ϵ) [14], i.e.,

$$\alpha(\epsilon, \theta) = 1 - \frac{1}{2} \left[\frac{1 + (1 - \epsilon \cos \theta)^2}{1 + (1 + \epsilon \cos \theta)^2} + \frac{\epsilon^2 - 2\epsilon \cos \theta + 2 \cos^2 \theta}{\epsilon^2 + 2\epsilon \cos \theta + 2 \cos^2 \theta} \right] \quad (9)$$

$$\epsilon = \frac{2\epsilon_2}{\epsilon_{1+} \left[\epsilon_1^2 + (\sigma_{st}/\omega\epsilon_0)^2 \right]^{1/2}} \quad (10)$$

It is noted that the value of ϵ is related to the electrical conductance per unit depth of the metal σ_{st} , and ϵ_1 and ϵ_2 donate the real parts of the dielectric constants of the metal and plasma, respectively. Moreover, ϵ_0 is the permittivity of a vacuum and ω represents the angular frequency of the laser beam.

2.2. Boundary and initial conditions

The boundary conditions in the present model are complex due to the many complicated physical phenomena at work in the laser welding process. As shown in Fig. 1, the top and bottom surfaces of the computational domain (ABCD and EFGH) are assigned outflow conditions and are assumed to be subject to heat conduction, heat convection, heat radiation, and evaporation effects. The thermal boundary for both surfaces is thus formulated as [14];

$$k \frac{\partial T}{\partial n} = -h_c(T - T_0) - \epsilon_r \sigma (T^4 - T_0^4) - \rho V_{evap} T \quad (11)$$

where $\vec{n}$ is the normal vector of the surface, h_c is the convection coefficient, T_0 is the environmental temperature, ϵ_r is the radiation coefficient, σ is the Stefan-Boltzmann constant, and V_{evap} is the free surface recession speed due to evaporation. It is noted that the evaporation recession speed, V_{evap} , can be calculated from the velocity jump across the Knudsen layer if the surface recession speed due to the evaporating mass flux is F_e , then the general form of the level set equation becomes [15];

$$\frac{\partial \varphi}{\partial t} + F_{lv} |\nabla \varphi| = 0 \quad (12)$$

$$F_{lv} = F_e + n \cdot U(x, y, z, t) \quad (13)$$

$$n = \frac{\nabla \varphi}{|\nabla \varphi|} \quad (14)$$

where $\varphi(x, y)$ is the zero-level set of the surface of interest, F_{lv} is the speed function, $U(x, y, z, t)$ is convection velocity, and n is the surface normal. In addition, the side surfaces (ABFE, BCGF, DCGH, and ADHE in Fig. 1) are assigned continuative conditions, and the thermal boundaries are thus given as

$$k \frac{\partial T}{\partial n} = -h_c(T - T_0) - \epsilon_r \sigma (T^4 - T_0^4) \quad (15)$$

For all of the simulations, the environmental temperature is assumed to be 298 K.

2.3. Verification of simulation model

The validity of the simulation model was confirmed by comparing the simulation results for the melt pool dimensions (depth and width) with the experimental results for representative values of the welding conditions. In performing the simulations, all of the parameters in Eqs. (1)–(15), with the exception of the electrical conductance (ϵ), were obtained directly from the literature. The related values are listed in

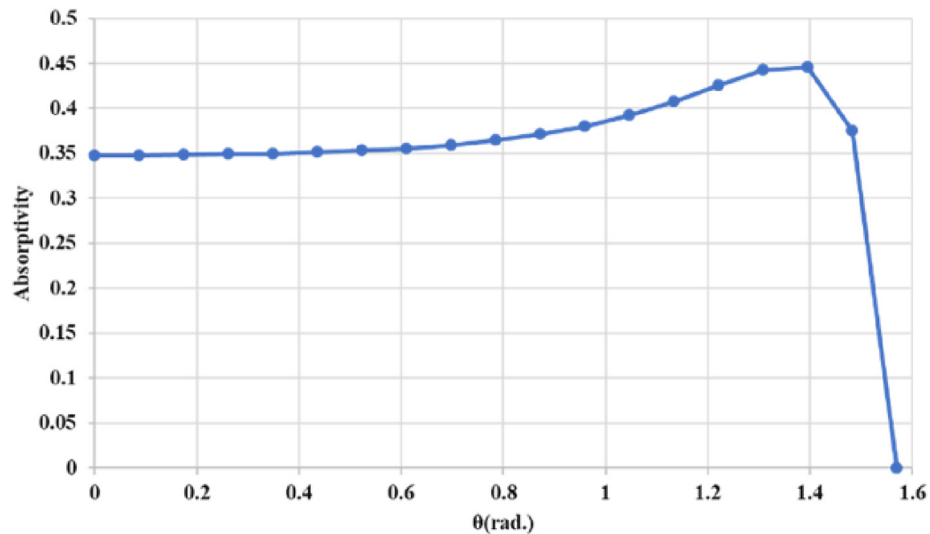


Fig. 2 – Fresnel absorptivity as function of angle between laser beam and free surface of keyhole. (Note that electrical conductance is $\varepsilon = 0.25$).

Table 1, where the workpiece material is assumed to be SS316 stainless steel. In general, the value of ε depends on both the workpiece material and the laser type. For example, Ducharme et al. [16] showed that a value of $\varepsilon = 0.08$ provided a good fit between the simulation results and experimental results for the melt pool dimension in the laser welding of steel using a CO₂ laser. However, Cho and Na [14] showed that a higher value of $\varepsilon = 0.25$ was required for the pulsed Nd:YAG laser processing of mild steel. In the present study, the welding trials were performed using a continuous wave fiber laser with a power of 100–3000 W and scanning speed of 10–160 mm/s. The results of a preliminary investigation showed that a value of $\varepsilon = 0.25$ yielded the best fit between the simulation results for the melt pool dimensions and the experimental results. Consequently, the Fresnel absorptivity in Eq. (10) was modeled using the function shown in Fig. 2, where θ is the angle between the laser beam and the free surface of the keyhole. The mesh size in modeling is an important design factor and is carefully considered for achieving high performance in Flow3D software. As depicted in Fig. 3, a finer mesh size of 40 μm was selected for the welding region, while a coarser mesh of 80 μm was chosen for the boundary walls. The total number of cells in the simulation block was 413,665.

Experimental welding trials were performed using three settings of the laser power (P) and scanning speed (V), namely $P: 2480 \text{ W}/V: 40 \text{ mm/s}$; $P: 2040 \text{ W}/V: 118 \text{ mm/s}$, and $P: 976 \text{ W}/V: 118 \text{ mm/s}$. Note that the parameter settings were deliberately chosen in such a way as to induce three different welding conditions: incomplete fusion, non-penetration, and full penetration, respectively. Following the welding trials, the weldments were cut, polished and etched, and the width and depth of the melt pool were measured using a microscope. Table 2 compares the simulation and experimental results for the melt pool dimensions in the three cases. (Note that the labels H, A and I in the left-hand column of the table indicate the relative positions of the welding parameters within the

feasible design space, as shown later in Fig. 12 in Sub-section 3.5.) It is seen that the maximum error between the two sets of results is just 16.84%. Thus, the basic validity of the CFD simulation model is confirmed. Fig. 4 compares the simulation results and experimental results for the cross-sections of the three melt-pools. Note that the red zone in each figure corresponds to a temperature of more than 1723 K (the melting point of SS316 [17] and hence indicates the melt pool. It is seen that a reasonable qualitative agreement exists between the two sets of results for each of the considered welding statuses (incomplete fusion, non-penetration, and full penetration). Thus, the basic validity of the thermal-fluid model is further confirmed.

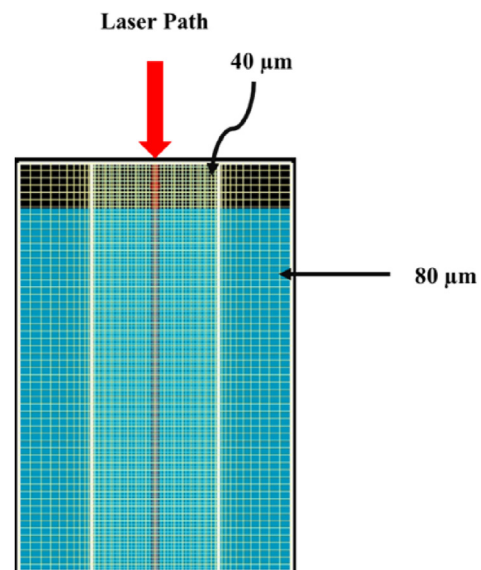


Fig. 3 – Finer and coarser mesh sizes from the top view of a scanning track.

Table 2 – Comparison of simulation and experimental results for melt pool dimensions.

Parameter	Width in simulation (mm)	Width in experiment (mm)	Error (%)	Depth in simulation (mm)	Depth in experiment (mm)	Error (%)
A (2480 W, 40 mm/s)	1.07	1.02	5.10	4.00	4.00	0.00
H (2040 W, 118 mm/s)	0.60	0.67	10.45	3.16	3.80	16.84
I (976 W, 119 mm/s)	0.50	0.55	9.09	1.50	1.44	4.17

According to the simulation results, the peak temperature exceeds the evaporation temperature of SS316 (3200 K [17]). Consequently, it is reasonable to assume that in the corresponding experiments, evaporation of the molten metal took place, which produced a recoil pressure and increased the melt pool depth (keyhole) and melt pool width. The good agreement between the simulation results and experimental results presented in Fig. 4(a, b and c) justifies the inclusion of the recoil pressure (Eq. (8)) and Fresnel absorptivity (Eq. (10)) in the proposed high-fidelity simulation model.

3. Optimization of welding parameters

3.1. Surrogate modeling

The present study considers the problem of identifying the processing parameters (i.e., the laser power and scanning speed) which optimize the quality of SS316 lap-joints fabricated using a laser welding technique. One possible approach for

identifying the optimal parameters is simply to use the validated CFD model described above to predict the melt pool formation for every feasible combination of the processing conditions within the design space (laser power: 100–3000 W, scanning speed: 10–160 mm/s). However, each simulation run takes on average around 10 h to complete when using an AMD Ryzen Threadripper 3970X 32-Core processor (CPU), and hence such an approach is clearly infeasible. Consequently, some form of sampling method [20] is required to minimize the number of input parameters which must be explored. Accordingly, the present study adopts the surrogate modeling approach shown in Fig. 5 to select appropriate values of the laser power and scanning speed for simulation purposes and to then predict the melt-pool dimensions and cooling rate for any combination of the two parameters in the feasible design space.

As shown in Fig. 5, the surrogate modeling process commences by selecting representative values of the laser power and scanning speed from the feasible design space using the sphere packing design method [19], which maps the design space with non-overlapping spheres of a pre-defined diameter

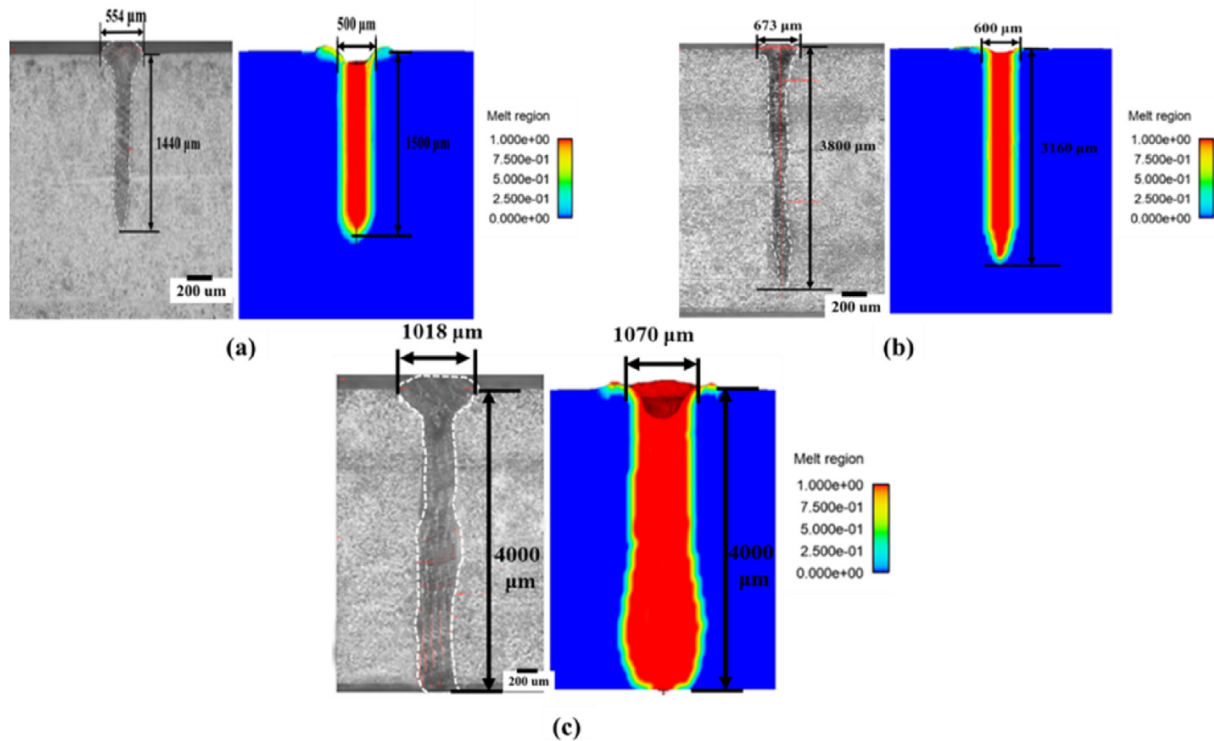


Fig. 4 – Comparison of simulation and experimental results for melt pool cross-sections, (a) $P = 976$ W, $V = 118$ mm/s; (b) $P = 2040$ W, $V = 118$ mm/s (c) $P = 2480$ W, $V = 40$ mm/s.

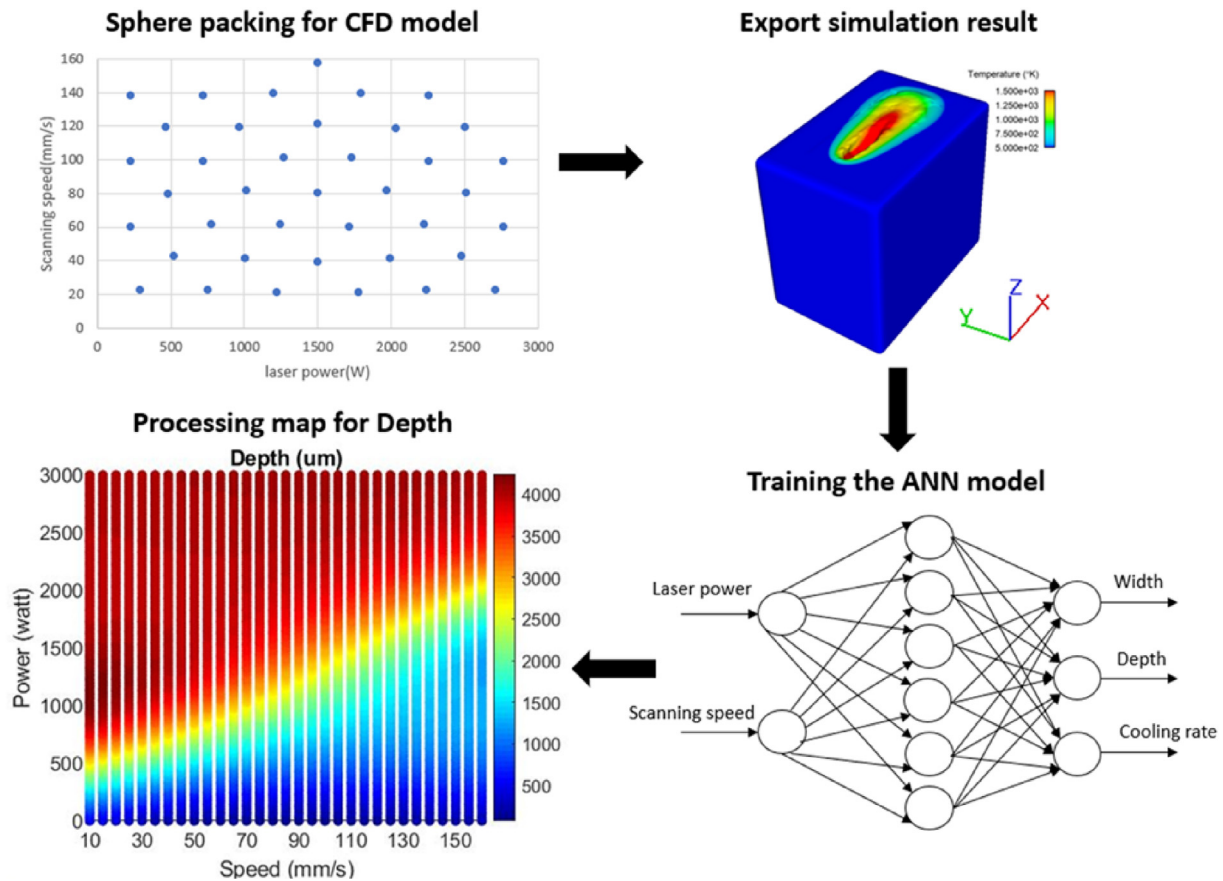


Fig. 5 – Schematic illustration of surrogate modeling approach.

and then selects the center point of each sphere as a sampling point. Fig. 6 illustrates the results obtained from the sphere packing algorithm for the present case where the scanning speed (x-axis) extends from 10–160 mm/s and the laser power (y-axis) extends from 100–3000 W. Note that in accordance with the sphere packing design method, the two axes are both normalized to the range of 0–1. Moreover, in the present

implementation, the packing sphere diameter was set as 0.15. As shown on the right-hand side of Fig. 6, a total of 39 representative combinations of the laser power and scanning speed were thus chosen as inputs for the CFD model described in Section 2.

For each selected set of processing conditions, the CFD model was used to predict the melt pool width, melt pool

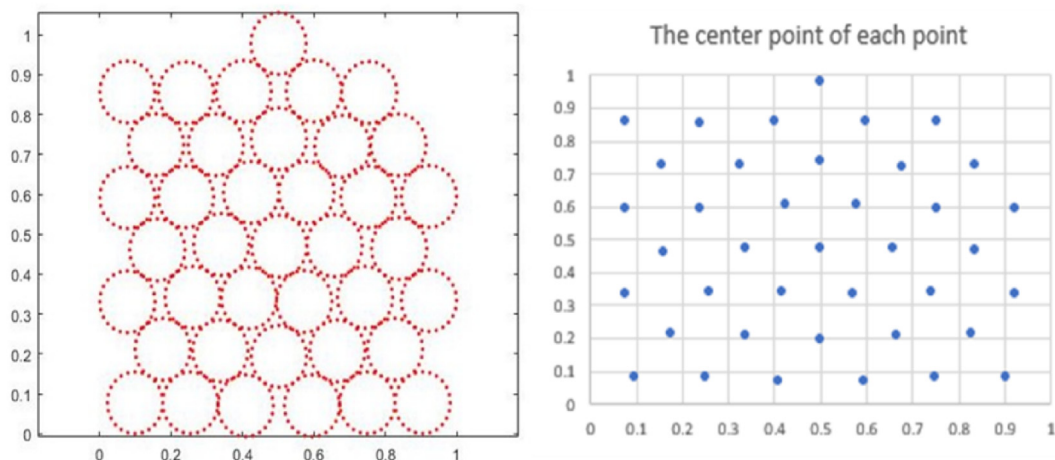


Fig. 6 – Sphere packing design results.

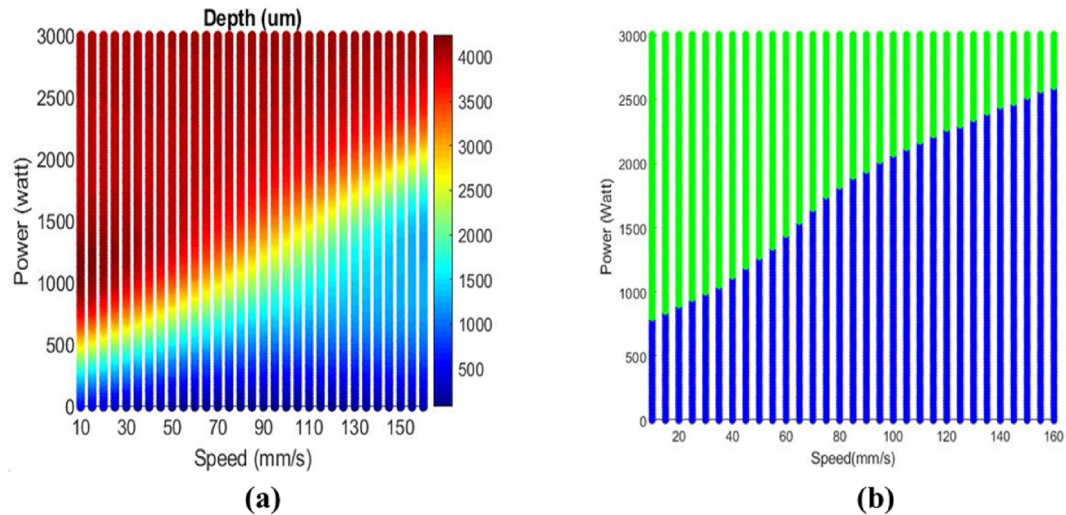


Fig. 7 – Processing and screen maps for (a) processing map for melt pool depth (unit: μm); (b) screened processing map in which green region shows melt pool depth greater than 3.5 mm and blue region shows melt pool depth less than 3.5 mm.

depth, and cooling rate, as shown in the top-right schematic in Fig. 5. However, the simulation results show the predicted welding outcome for only a very limited number of welding conditions. Thus, there is no guarantee that they include the welding parameters required to optimize the quality of the lap-joint. In other words, to identify the optimal parameters, it is still necessary to explore the entire design space in some way. In the present study, this was achieved utilizing an artificial neural network (ANN) model.

ANNs have been widely used in computer vision and data mining applications in recent years and provide a highly effective method for analyzing problems associated with inexactitude and fuzzy information [19]. In the present study, an ANN with two input nodes (laser power and scanning speed) and three output nodes (the melt pool width, melt pool depth, and cooling rate) was constructed using R programming software [19]. The model was trained

using the simulation results obtained for the 39 parameter combinations selected using the sphere packing design method. The trained model was then used to predict the melt pool width, melt pool depth, and cooling rate for 3752 sampling points uniformly distributed within the design space with a laser power interval of 25 W and scanning speed interval of 5 mm/s. The results obtained for each output of the ANN model were plotted in the form of a processing map (see bottom-left schematic in Fig. 5 for illustration purposes).

Finally, the processing maps obtained for the melt pool depth, melt pool width, and cooling rate, respectively, were screened with appropriate quality criteria to determine the laser power and scanning speed which minimized the residual stress and porosity of the lap-joint weld and improved the mechanical properties. The details of the screening process are described in the following sub-sections.

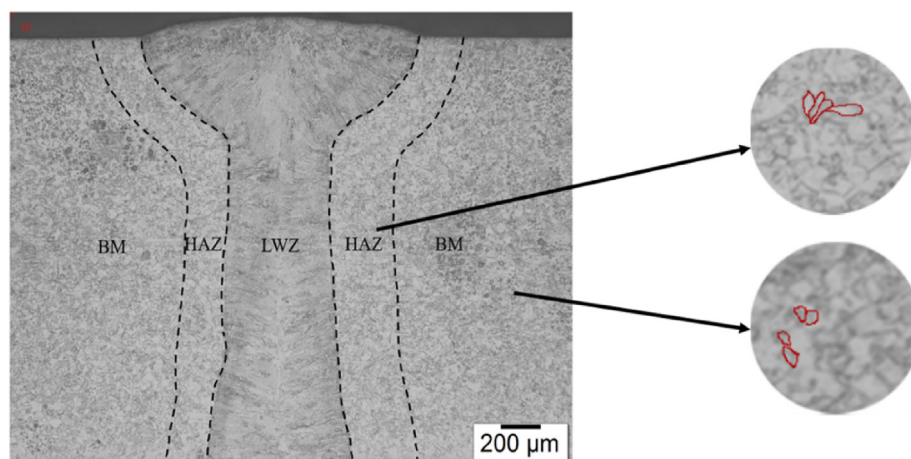


Fig. 8 – Cross-sectional optical microscope image of lap-joint produced using laser power of 1785 W and scanning speed of 20 mm/s.

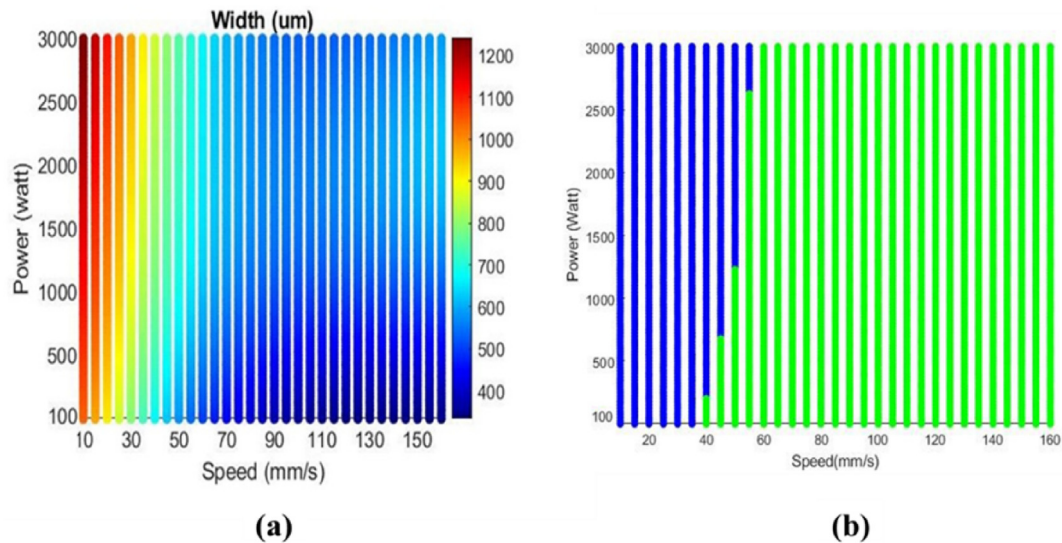


Fig. 9 – Processing and screen maps for (a) processing map for melt pool width (unit: μm); (b) screened processing map in which green region shows melt pool width less than 0.7 mm and blue region shows melt pool width greater than 0.7 mm.

3.2. Melt pool depth criterion

To join the two SS316 plates, the melt pool depth should be at least 2 mm. However, if the energy density is too high, the melt pool depth may exceed the total weldment thickness (4 mm). Moreover, in lap-joint welding, excessive porosity seriously degrades the weld strength. Hence, an appropriate selection of the welding parameters is essential. In deep-penetration welding, vapor bubbles are readily formed at the bottom of the keyhole due to the effects of intense evaporation [21,22]. If these bubbles are not voided from the keyhole during the welding process, they become trapped within the solidified microstructure as porosity [23]. However, if the processing conditions are controlled in such a way that the melt pool depth reaches full penetration (4 mm in the present case), the air in the trapped voids is exhausted along the bottom of the workpiece before solidification takes place.

Fig. 7(a) shows the processing map obtained from the trained ANN for the melt pool depth given 3752 combinations of the laser power and scanning speed within the feasible design space. Based on the error margin between the simulation results and experimental results for the melt pool depth (16.84%, see Table 2), the criterion for screening the processing map was set as a minimum melt pool depth of 3.5 mm. The corresponding results are shown in Fig. 7(b), in which the parameter settings which achieve a full penetration depth of 3.5–4 mm, and hence minimize porosity formation, are shown in the green region.

3.3. Melt pool width criterion

In general, the cross-sectional microstructure of a lap-joint can be divided into three main regions: the laser welding zone (LWZ), the heat affected zone (HAZ), and the base metal (BM)

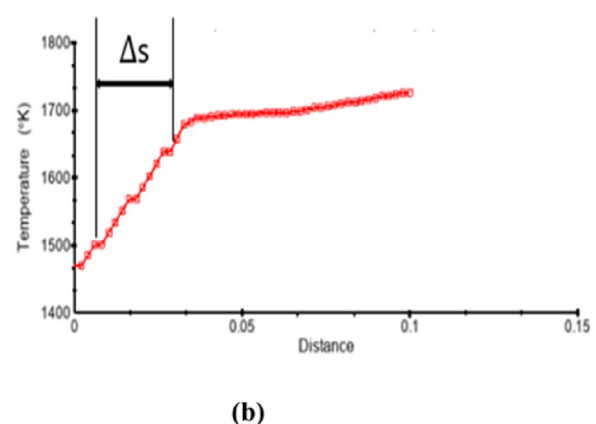
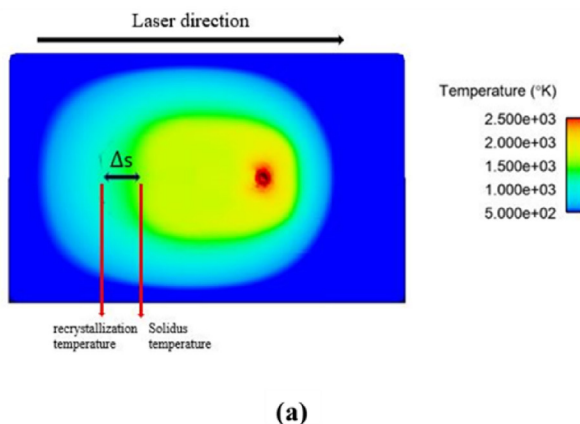


Fig. 10 – Simulation results for (a) distance between solidus temperature and recrystallization temperature; (b) temperature distribution along scanning direction.

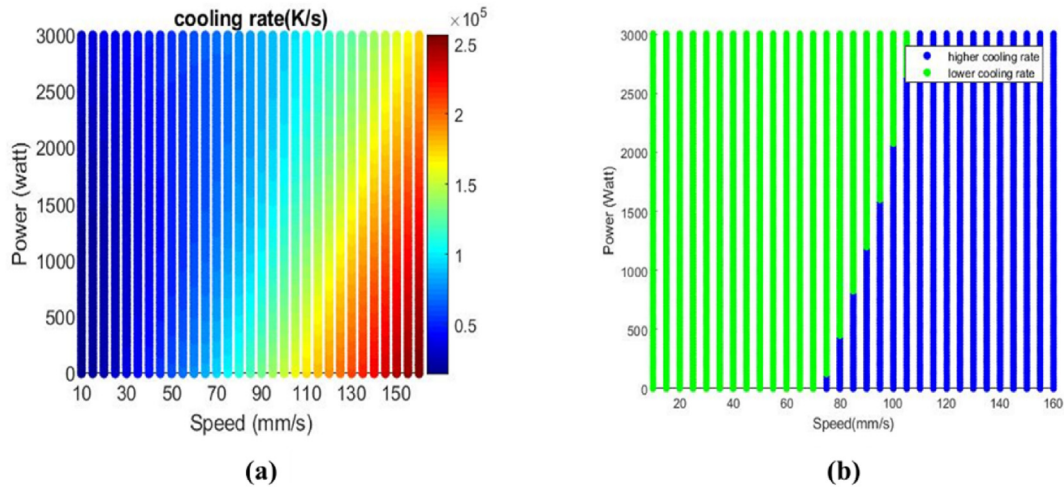


Fig. 11 – Processing and screen maps for (a) processing map for cooling rate (unit: K/s); (b) screened processing map in which green region shows cooling rate lower than 1×10^5 K/s and blue region shows cooling rate greater than 1×10^5 K/s.

zone (see Fig. 8). The LWZ has a keyhole profile with a narrow width and large depth due to the high energy intensity of the laser beam [24]. The BM region of the weld has a relatively coarse grain microstructure with a regular polygonal arrangement of the grains. However, the LWZ contains a fine grain structure with columnar crystals and dendrites with different orientations. The high magnification morphology of the center of the laser welding zone indicates that the crystal solidified microstructure is dendrite and the dendrite microstructures are a larger size in the middle of the weld [25]. A detailed inspection of Fig. 8 shows that the grain size in the HAZ is obviously larger than that in the BM. The coarse grains not only lead to a local degradation of the mechanical properties of the weld, but also impair the corrosion resistance [26]. Consequently, the welding parameters should be set in such a way as to minimize the HAZ. The size of the HAZ is related to the width of the melt pool. Thus, in the lap-joint welding process, it is desirable to

find the welding parameters which minimize the melt pool width and hence reduce the size of the HAZ.

Fig. 9(a) shows the processing map obtained from the trained ANN for the melt pool width. In practice, there is no specific value of the melt pool width which minimizes the size of the HAZ and hence improves the mechanical properties and corrosion resistance of the weld. Thus, in screening the processing map shown in Fig. 9(a), the quality criterion was selected simply as the mean value (0.7 mm) of the melt pool width over all the considered parameter combinations. The screening results are shown in Fig. 9(b), in which the green region indicates the favored processing region with a melt pool width of less than 0.7 mm.

3.4. Cooling rate criterion

In the laser welding process, the rate at which the solid phase cools from the solidus temperature to the recrystallization temperature has a critical effect on both the grain size and the residual stress [23]. Generally speaking, a higher cooling rate gives rise to a larger residual stress, and thus the processing parameters should ideally be designed in such a way as to reduce the cooling rate. The solid cooling rate can be defined as shown in Eq. (16), details can be found in [23].

$$\text{Solid cooling rate} = \frac{T_{\text{solidus}} - T_{\text{recrystallize}}}{\Delta t} = \frac{T_{\text{solidus}} - T_{\text{recrystallize}}}{|d_{\text{solidus}} - d_{\text{recrystallize}}| / v_s} \quad (16)$$

where T_{solidus} is the solidus temperature, $T_{\text{recrystallize}}$ is the recrystallization temperature, and Δt is determined by dividing the distance between the solidus temperature and the recrystallization temperature by the scan speed. Note that the solidus temperature and recrystallization temperature given in Fig. 10 are for stainless steel 316. The corresponding values for the SS316 workpiece material considered in the present study are 1658 K and 1223 K, respectively.

As shown in Fig. 11, in the present simulations, the temperature was evaluated along the scanning direction for each of

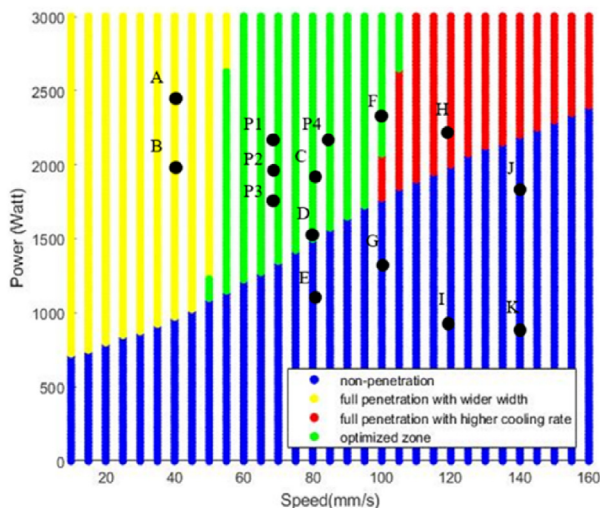
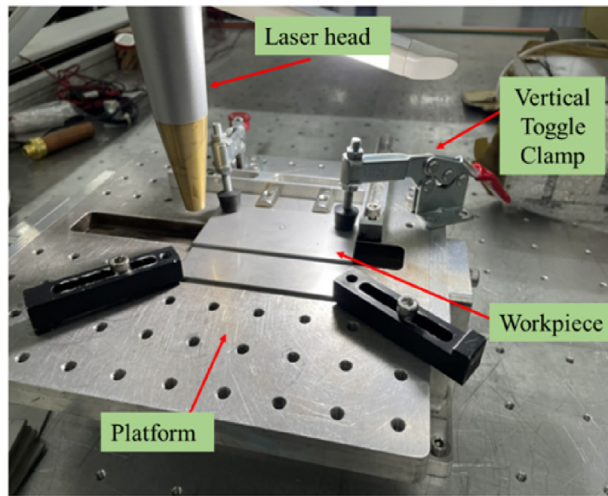


Fig. 12 – Final screened processing map.

Table 3 – Composition of SS316 plates (wt.%).

C	Si	Mn	P	S	Cr	Mo	N	Ni	Fe
0.08	0.75	2.00	0.045	0.03	16.0–18.0	2.0–3.0	0.1	10.0–14.0	Balance

**Fig. 13 – Experimental setup for laser welding trials.**

the 39 representative processing conditions, and the cooling rate was determined by dividing the scanning distance, ΔS , between the solidus temperature and recrystallization temperature by the corresponding scanning speed. The 39 values of the cooling rate were then used to train the ANN, as described earlier in Section 3.1. Fig. 11(a) shows the processing map obtained from the trained model for the cooling rates of the 3752 parameter combinations within the feasible design space. It is seen that the cooling rate varies in the range of 0.15×10^5 to

2.5×10^5 K/s. As for the cooling rate, there is no specific value of the cooling rate which minimizes the residual stress within the solidified microstructure. Thus, in screening the processing map, the quality criterion for the cooling rate was chosen simply as the mean value of the cooling rate over all 3752 parameter combinations, i.e., 1×10^5 K/s. The screening results are presented in Fig. 11(b), where the green region indicates the favored processing region with a cooling rate of less than 1×10^5 K/s.

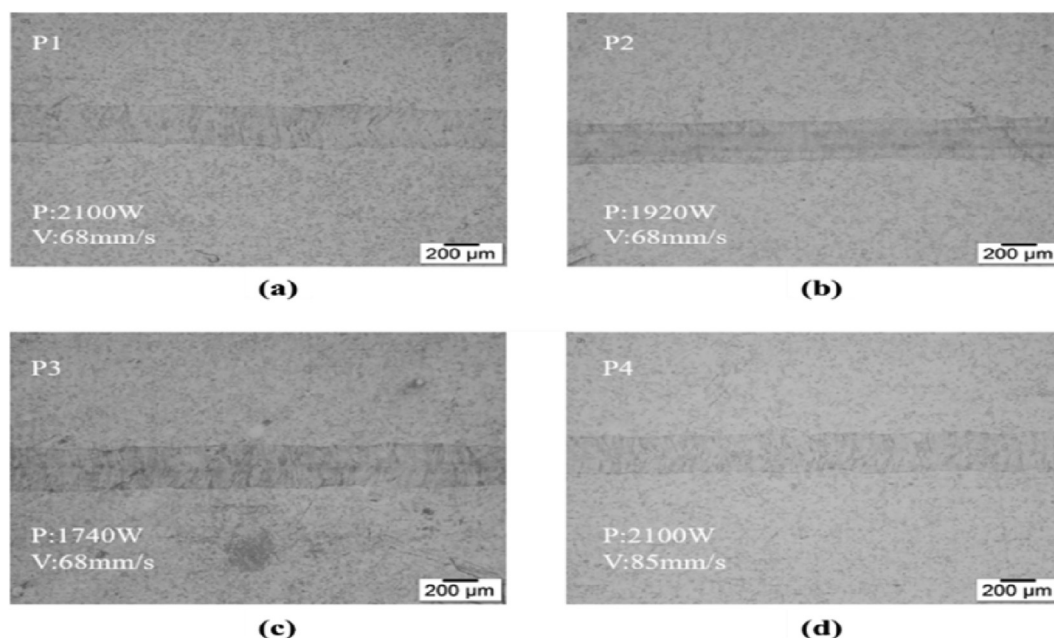
3.5. Final processing map

Fig. 12 shows the final processing map obtained by superimposing the individual maps in Figs. 7, 9 and 11 for the melt pool depth, melt pool width, and cooling rate, respectively. Finally, the green region indicates the optimal processing region, in which the processing conditions are expected to yield a SS316 lap-joint with low porosity, a reduced HAZ size, and a lower residual stress.

4. Results and discussions

4.1. Comparison of experimental and simulation results

The feasibility of the proposed parameter optimization method was confirmed by performing experimental lap-joint welding trials using SS316 plates with the dimensions shown in Fig. 1 and the chemical composition shown in Table 3. It is noted that sampling procedure [18] for the optimization method is

**Fig. 14 – Underside views of lap-joints produced using optimal processing conditions at Points P1 – P4 in final screened processing map (Fig. 12).**

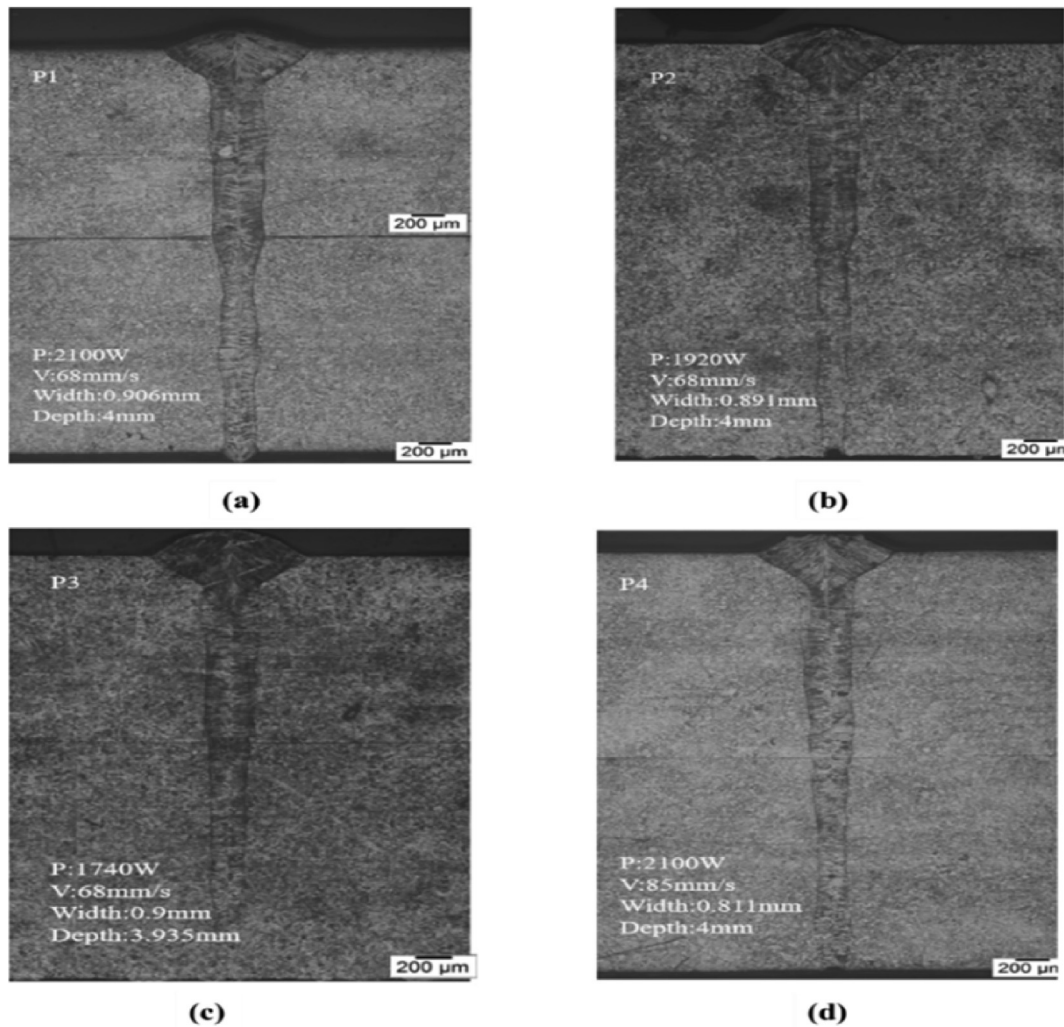


Fig. 15 – Cross-sectional views of lap-joints produced using optimal processing conditions at Points P1–P4 in final screened processing map (Fig. 12).

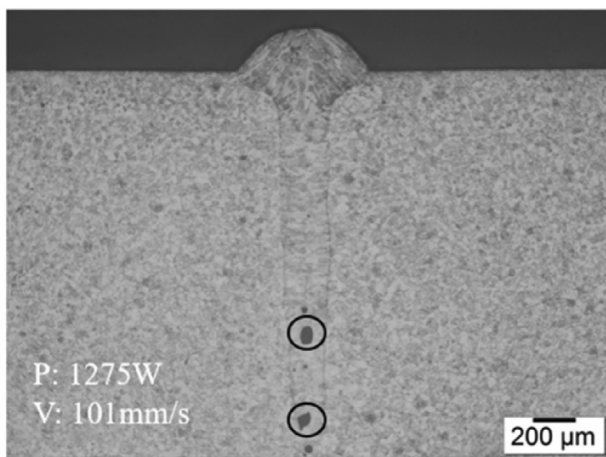


Fig. 16 – Cross-sectional view of lap-joint produced using non-optimal processing conditions at Point G in final screened processing map in Fig. 12.

necessary to minimize the number of input parameters to examine. The trials were performed using a fiber laser welding system (Model: BW250WF110, Jumbo, Taiwan) fitted with a RAYTOOLS welding head (see Fig. 13). The laser source had a central wavelength of 1064 nm and a spot diameter of 50 µm at the focal plane. For each selected set of processing parameters, the lap-joint was fabricated using a single-track scanning strategy. Argon gas with a flow rate of 20 L/min was used as the shielding gas in every case. After the trials, the samples were ground using abrasive paper, polished by polishing powder, and finally etched with corrosive liquid. The abrasive paper is from 240 grit to 3000 grit. The polishing liquid is composed of 1 µm silicon oxide paste. The etchant is composed of 30 ml of distilled water, 20 ml HCL and 15 ml HNO₃. It is used at room temperature for 120s. The cross-sectional dimensions of the weld pool were then measured under an optical microscope.

It is noted that Fig. 12 indicates the various processing conditions considered in the welding trials. Points A, H and I indicate the parameter settings used to perform the initial

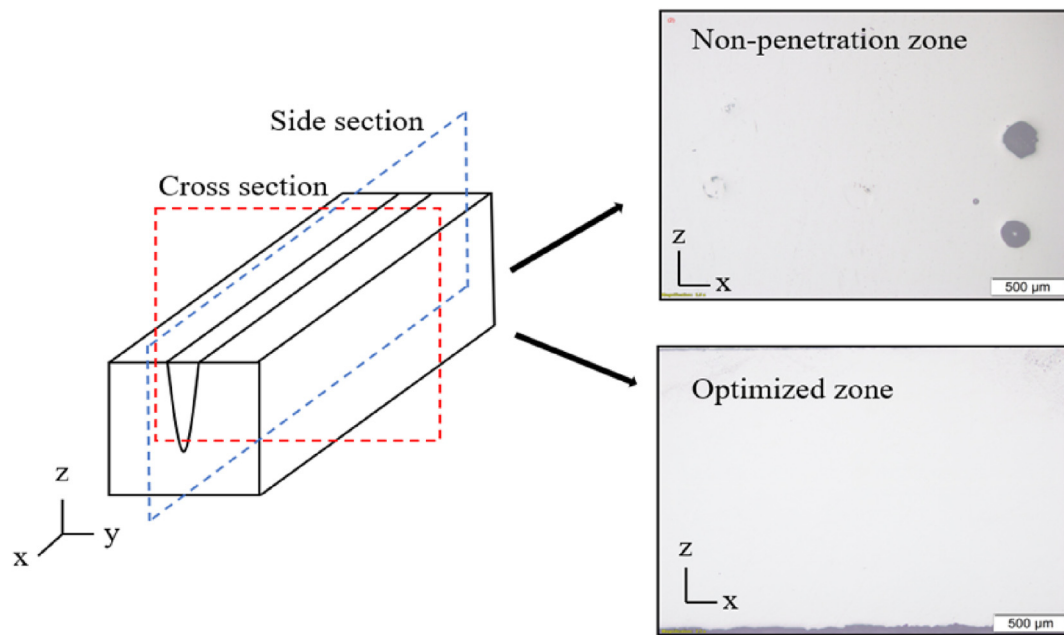


Fig. 17 – Side-sectional views of lap-joints in non-penetration zone and optimized zone.

verification of the CFD model in Section 2.3 (see Table 2). In addition to these three points, welding trials were performed using eight further sets of processing parameters distributed throughout the feasible design space (i.e., Points B - G and J - K). Finally, further trials were also performed using four sets of processing parameters chosen from the optimal green region of the processing map, namely Point P1 ($P = 2100$ W, $V = 68$ mm/s), Point P2 ($P = 1920$ W, $V = 68$ mm/s), Point P3 ($P = 1740$ W, $V = 68$ mm/s), and Point P4 ($P = 2100$ W, $V = 85$ mm/s). Fig. 14 presents optical microscope images of the underside of the lap-joints produced using the processing parameters at Points P1 – P4. Fig. 15 shows the corresponding

cross-sectional views. The images presented in Fig. 14 confirm that almost no pores are present in the lower region of the melt pool. Moreover, the welding seam has a consistent width of around $200\ \mu\text{m}$ in every case.

By contrast, the cross-sectional image in Fig. 16, corresponding to Point G in Fig. 11 with a melt-pool depth of less than 3.5 mm, shows significant porosity in the lower keyhole region. In order to effectively observe and quantify the amount of porosity, the side-sectional images were used to calculate the porosity through a commercial software “ImageJ”. Fig. 17 shows the side-sectional views of lap-joints in the porosity distribution in the optimized zone and non-

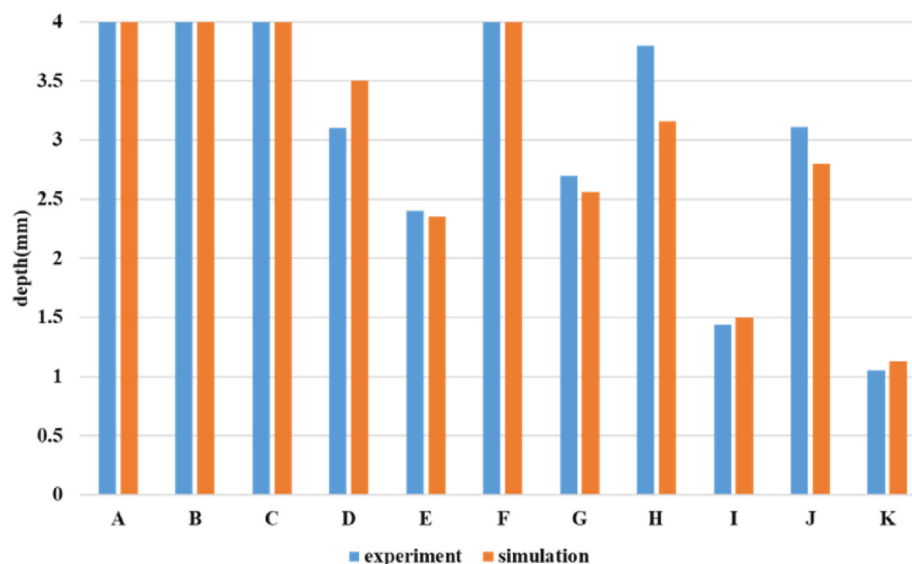


Fig. 18 – Comparison of experimental and simulation results for melt pool depth.

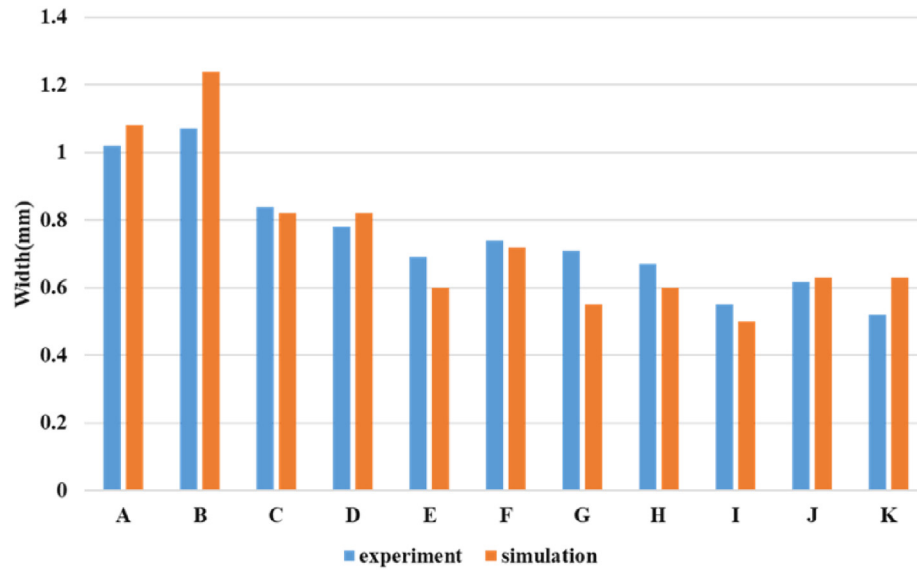


Fig. 19 – Comparison of experimental and simulation results for melt pool width.

penetration zone, respectively. It indicates that the parameters in the optimized zone have less porosity compared to the parameters in non-penetration zone. Obviously, the porosity is reduced from 1.22% in non-penetration zone to 0.21% in optimized zone. In general, the findings illustrated in Figs. 14 and 15 support the optimal processing conditions within the green region depicted in Fig. 12, thereby providing evidence for the efficiency of the proposed optimization framework. In addition, Fig. 16 displays noticeable pores resulting from the use of non-optimal parameters referred to as Point G in the final processing map, which is indicated by the blue region.

Figs. 18 and 19 compare the experimental and simulation results for the melt pool depth and melt pool width, respectively, of the lap-joints produced using the processing conditions at Points A to K in Fig. 12. The average errors between the two sets of results for the melt pool depth and melt pool width are found to be just 5.34% and 10%, respectively. Thus, the basic validity of the CFD simulation model is once again confirmed. It was reported in Refs. [27,28] that simulation models which ignore the recoil pressure tend to underestimate the melt pool dimensions. Moreover, the deviation of the simulation results from the experimental observations tends to increase as the energy density rises. However, no such tendency is observed in the present results due to the higher

fidelity of the simulation model, which takes account not only of the recoil pressure (Eq. (8)), but also the Fresnel absorptivity (Eq. (10)).

4.2. Mechanical properties of lap joint in tensile tests

To evaluate the mechanical properties of the lap-joints produced using the optimal processing conditions, the tensile test specimens with the dimensions and geometry shown in Fig. 20 were fabricated using the parameters associated with Points P1, P2, P3, and P4 in the green region of Fig. 12. The tensile strengths of the specimens were measured in accordance with the ASTM E8 standard [29] on a universal testing machine (HT-2402, Hung Ta, Taichung, Taiwan), as shown in Fig. 21. Each test was carried out at room temperature 25 °C with a load-line displacement speed of 1.2 mm/min and a strain rate of 0.5.

For lap-joint specimens, the main mechanical property of interest is the ultimate shear strength. However, the universal testing machine is capable of measuring only the yield tensile strength and ultimate tensile strength of the samples. Accordingly, following the tensile tests, the interface width (IW) between the two plates of the lap-joint weld (see Fig. 20) was measured using an optical microscope and the ultimate

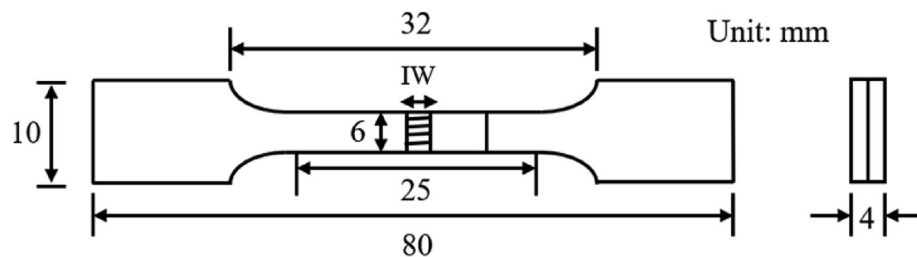


Fig. 20 – Dimensions and geometry of tensile test specimens.



Fig. 21 – Universal tensile test machine used to evaluate ultimate tensile strength of lap-joint samples.

tensile strength values obtained from the universal testing machine (see Fig. 21) were converted into corresponding ultimate shear strength value accordingly (i.e., loading divided by the interface area). The ultimate tensile strength of the base metal (SS316) was found to be 687.5 MPa. It is also worth noting that unlike ultimate tensile strength and yield tensile strength, the standard value of shear strength has not been published. However, 60% of ultimate tensile strength in base metal workpiece is commonly used as an approximation for the ultimate shear strength. Thus, in accordance with common practice, the ultimate shear strength of the base metal was estimated to be 412.5 MPa (i.e., 60% of the measured ultimate tensile strength). Table 4 compares the ultimate shear strength values of the SS316 base metal and four tensile specimens fabricated using the optimal processing conditions. It is seen that all four tensile test specimens have a higher ultimate shear strength than the base metal. The specimen fabricated using a laser power of 1740 W and scanning speed of 68 mm/s has the highest ultimate shear strength of 545.77 MPa, which is around 32% higher than that of the base metal. According to the ASTM E8 standard, the ultimate shear strength of lap-joint samples should exceed 80% of the ultimate shear strength of the base metal (i.e., 330 MPa for the present SS316 alloy). As shown in Tables 4 and it is evident that the ultimate shear strengths of all four lap joints exceed the anticipated value by a considerable amount. This outcome offers conclusive proof of the effectiveness of the optimized parameters in improving the mechanical characteristics of the welded microstructure and reducing residual stress.

Table 4 – Ultimate shear strengths of SS316 base metal and tensile test samples.

Parameter	Ultimate shear strength (MPa)
Base metal	412.5
P1 (2100 W/68 mm/s)	455.88
P2 (1920 W/68 mm/s)	398.27
P3 (1740 W/68 mm/s)	545.77
P4 (2100 W/85 mm/s)	398.47

5. Conclusions and future work

This study is commenced by constructing a high-fidelity CFD model to evaluate the effects of the laser welding parameters (i.e., the laser power and scanning speed) on the melt pool dimensions and cooling rate of SS316 lap-joints. To ensure the accuracy of the simulation results, the model took account of all the main physics affecting the flow field of the weld pool, including the Marangoni force, the recoil pressure, and the vapor drag force. The major contributions and findings of the present study can be summarized as follows.

- The artificial neural network (ANN) used in the study helped predict the melt pool dimensions and cooling rate for 3752 possible combinations of laser power and scanning speed within the feasible design space.
- The validity of the simulation model was confirmed by comparing the simulation results for the melt pool dimensions (width and depth) with the experimental data. It was shown that the average discrepancies between the experimental and simulation results for the melt pool depth and melt pool width were just 5.34% and 10%, respectively.
- The experimental observations have shown that lap welds made using optimal processing conditions achieve the full weld penetration depth, resulting in minimal porosity density in the solidified weld pool reduced from 1.22% in a non-penetration zone to 0.21% in an optimized zone. Furthermore, the tensile test results showed that the ultimate shear strength of the lap-joints exceeded that of the base metal by up to 32%.
- The study found that the processing parameters that minimized porosity and residual stress within the welded lap-joint and enhanced the mechanical properties were determined based on three quality criteria, namely, melt-pool depth, melt-pool width, and cooling rate.

It is noted that the present study considered only a simple single-track scanning strategy; however, different scanning strategies, e.g., Hilbert curve, triangle weave, and wobbling,

result in different melt-pool dimensions and HAZ widths, and therefore affect the choice of optimal processing conditions. Thus, future studies should extend the CFD simulation model and ANN framework proposed in this study to more complex scanning strategies. Furthermore, the present study assumed that the focal plane of the fiber laser was coincident with the surface of the SS316 workpiece. However, while this assumption simplifies the modeling process, the effect of focus plane is also an important issue for laser welding. Hence, it should extend the CFD model with different focus plane. Moreover, future studies should also investigate the effect of different environment pressure on the melt pool dimension. Moving forward, the further analyses will be conducted to precisely quantify the effects of keyhole instability, collapse, and bubble behavior on porosity formation. Specifically, different shielding gas configurations will be tested to identify those factors that can minimize porosity.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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